

Unit 5 Notes

Similarity

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 5 Notes: Similarity

Big goal

Similarity means same shape, not necessarily same size. Dilations create scaled copies, and similarity proofs depend on equal corresponding angles and proportional corresponding side lengths.

1. Dilations

A dilation has a center and a scale factor k . It sends a point along the ray from the center through the point. If $|k| > 1$, the image is larger. If $0 < |k| < 1$, the image is smaller. A negative scale factor also reverses through the center.

For a dilation centered at the origin:

$$(x, y) \mapsto (kx, ky).$$

For center (a, b) , a point (x, y) maps to

$$(a + k(x - a), b + k(y - b)).$$

2. Properties of dilations

A dilation sends a line not passing through the center to a parallel line. A line passing through the center stays on the same line. Segment lengths are multiplied by $|k|$, angle measures are preserved, and parallelism is preserved.

3. Similar figures

Two figures are similar if all corresponding angles are congruent and all corresponding sides are proportional. A similarity transformation is a sequence of rigid motions and dilations.

Scale factor effects

If the length scale factor is k , then perimeter also scales by k , area scales by k^2 , and volume scales by k^3 for solids.

4. Triangle similarity theorems

Criterion	What proves triangles similar
AA	Two pairs of corresponding angles are congruent. The third pair follows automatically.
SAS Similarity	Two pairs of corresponding sides are proportional and the included angles are congruent.
SSS Similarity	All three pairs of corresponding sides are proportional.

5. Proportionality with parallel lines

If a line parallel to one side of a triangle intersects the other two sides, it divides those sides proportionally. Conversely, if a line divides two sides of a triangle proportionally, then the line is parallel to the third side.

Parallel-line proportion

If $DE \parallel BC$ in $\triangle ABC$, with D on AB and E on AC , then

$$\frac{AD}{DB} = \frac{AE}{EC} \quad \text{and} \quad \frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}.$$

6. Similar right triangles and the Pythagorean Theorem

Dropping an altitude from the right angle of a right triangle to the hypotenuse creates three similar right triangles. This similarity can prove $a^2 + b^2 = c^2$ and creates geometric mean relationships:

$$h^2 = pq, \quad a^2 = cp, \quad b^2 = cq,$$

where p and q are the two hypotenuse segments, h is the altitude, and $c = p + q$ is the hypotenuse.

7. Writing similarity proofs

A strong similarity proof identifies corresponding vertices. For example, writing $\triangle ABC \sim \triangle DEF$ means $A \leftrightarrow D$, $B \leftrightarrow E$, and $C \leftrightarrow F$. After similarity is proved, corresponding sides can be placed in proportions.

8. Worked examples**Example 1: Solve a proportion**

If $\triangle ABC \sim \triangle DEF$, and $AB = 6$, $DE = 9$, $BC = 10$, then

$$\frac{AB}{DE} = \frac{BC}{EF} \Rightarrow \frac{6}{9} = \frac{10}{EF}.$$

So $6EF = 90$, and $EF = 15$.

Example 2: Dilation about a point

Dilate $P(6, 9)$ by $k = \frac{1}{2}$ centered at $C(2, 1)$:

$$(2 + \frac{1}{2}(6 - 2), 1 + \frac{1}{2}(9 - 1)) = (4, 5).$$

Example 3: Shadows

A 6-ft person casts a 4-ft shadow. A tree casts a 30-ft shadow. Similar triangles give

$$\frac{6}{4} = \frac{h}{30} \Rightarrow 4h = 180 \Rightarrow h = 45.$$

The tree is 45 ft tall.

Mastery checklist

You are ready for Unit 5 when you can apply dilation rules, find scale factors, prove triangles similar by AA/SAS/SSS, use parallel-line proportions, solve similarity word problems, and connect similarity to the Pythagorean Theorem.